This is a review submitted to Mathematical Reviews/MathSciNet.

Reviewer Name: Sankarnarayanan, Brahadeesh

Mathematical Reviews/MathSciNet Reviewer Number: 169002

Address:
Department of Mathematics
Indian Institute of Technology Jodhpur
NH65, Nagaur Road, Karwar
Jodhpur Rajasthan, 342037
INDIA
briha1994@gmail.com

Author: Liu, Dan-Dan; Xu, Shou-Jun

Title: The planar Turán number of double star $S_{3,5}$.

MR Number: MR5105959

Primary classification: 05C35

Secondary classification(s): 05C10

Review text:

Given a graph H, the planar Turán number $ex_{\mathcal{P}}(n, H)$ is the maximum number of edges in an H-free planar graph on n vertices, where a graph is H-free if it contains no subgraph isomorphic to H. For positive integers k and l, the double star $S_{k,l}$ is the tree obtained by joining the centres of two vertex-disjoint stars with k and l leaves, respectively.

The study of planar Turán numbers was initiated by Dowden [J. Graph Theory **83** (2016), no. 3, 213–230; MR3549506]. Lan–Shi–Song [Electron. J. Combin. **26** (2019), Paper No. 2.11; MR3956449] established several sufficient conditions for a graph H to satisfy $ex_{\mathcal{P}}(n, H) = 3n - 6$, including the condition that H is K_4-free and $\Delta(H) \geq 7$. In particular, this equality holds for every double star $S_{k,l}$ with $l \geq 6$. The equality also holds when $k \geq 4$, since the graph $2K_1 + C_{n-2}$, obtained by joining two nonadjacent vertices to every vertex of a cycle on $n-2$ vertices, is an $S_{k,l}$-free planar triangulation. Consequently, among double stars $S_{k,l}$ with $2 \leq k \leq l$, only the cases $2 \leq k \leq 3$ and $l \leq 5$ require consideration. Ghosh–Győri–Paulos–Xiao ["Planar Turán number of double stars", Preprint (2021), arXiv:2110.10515] initiated a systematic study of these cases, determining the exact planar Turán numbers of $S_{2,2}$ and $S_{2,3}$, and upper bounds for $S_{2,4}$, $S_{2,5}$, $S_{3,3}$ and $S_{3,4}$. Subsequently, Xu–Hu–Zhang [Discrete Appl. Math. **358** (2024), 326–332; MR4778796] improved the upper bound for $S_{2,5}$. Further results were obtained by Xu–Zhou–Li–Yan ["Planar Turán number for balanced double stars", Preprint (2024), arXiv:2406.05758] for $S_{3,3}$, Xu–Shao–Zhou [Discrete Appl. Math. **369** (2025), 131–136; MR4881493] for $S_{2,5}$, Xu–Shao [Dis-

crete Math. **348** (2025), no. 11, Paper No. 114571; MR4908030] for $S_{2,4}$, and Xu–Zhang–Shao [AIMS Math. **10** (2025), no. 1, 1628–1644; MR4854366] for $S_{3,4}$. This leaves $S_{3,5}$ as the only remaining case in this family for which no corresponding nontrivial upper bound had been established.

The authors claim that

$$ex_{\mathcal{P}}(n, S_{3,5}) \leq \frac{23n}{8} - \frac{9}{2}$$

for every integer $n \geq 2$. They observe that the bound is tight for the icosahedral graph G_0, the 5-regular planar triangulation on 12 vertices with 30 edges.

However, the claimed upper bound is false. Let G_{13} be obtained from G_0 by inserting a new vertex into a triangular face and joining it to the three vertices of that face. The resulting graph is a planar triangulation on 13 vertices with 33 edges, having three vertices of degree six, nine of degree five, and one of degree three. Moreover, G_{13} is $S_{3,5}$-free. Indeed, the centre of the star with five leaves in any copy of $S_{3,5}$ must be one of the three degree-six vertices. If its adjacent centre has degree five, the two centres have at least two common neighbours, since G_{13} is a triangulation. If the adjacent centre also has degree six, they have three common neighbours. Thus, G_{13} is $S_{3,5}$-free, and Euler's bound for simple planar graphs gives

$$ex_{\mathcal{P}}(13, S_{3,5}) = 33 > \frac{23(13)}{8} - \frac{9}{2} = \frac{263}{8}.$$

The graph G_{13} also contradicts two intermediate claims in the proof. Claim 2.2 asserts the stated bound whenever an $S_{3,5}$-free planar graph has an edge xy with $d(x) + d(y) \leq 10$. But the new degree-three vertex in G_{13} is adjacent to three degree-six vertices, giving edges of degree sum nine. The same graph contradicts Claim 2.4, since the new vertex determines an edge cut of size three.

Consequently, the proposed upper bound does not hold in general, and the problem of determining $ex_{\mathcal{P}}(n, S_{3,5})$ remains open.